

Kotlin:

A Fun Language for Android App Development



Kotlin is a statically typed programming language for modern multi-platform applications. It is 100 per cent interoperable with Java. Existing libraries and frameworks in Java can directly be run on Kotlin, which also happens to be open source and freely distributed under the Apache licence.

Kotlin is a statically typed open source modern programming language targeting JVM, Android and even JavaScript. It is named after the Russian island called Kotlin, just as Java is named after the Indonesian island. It works everywhere where Java works – on the server side, the desktop, Android, and other places.

The biggest selling point of Kotlin is its 100 per cent interoperability with Java. So instead of throwing away or re-writing all your libraries in Kotlin, you can use all your existing Java libraries and frameworks directly with it. Due to this characteristic, you can gradually adopt Kotlin in your Java code base.

Kotlin has been created by JetBrains, the maker of IntelliJ, Android Studio and other amazing IDEs.

It is open source and has been released under the Apache licence.

Why you should use Kotlin

The biggest reason to learn Kotlin is that at Google I/O 17, Google announced first-class support for Kotlin for Android. So now you can expect full support from Google, like documentation to write Android apps in Kotlin.

Another reason is that it's an amazing language and many developers (especially Android developers like me) are in love with it and have fun writing code in it. People who are tired, or in some cases irritated with Java, will definitely find Kotlin concise and fun to work with.

Problems and issues with Java and the Android platform

Let's start with the current problems with Java that are usually faced by developers coming from other languages.

Java is extremely old. It's been around for more than 20 years. It doesn't have the modern higher order functions and features that newer languages like Swift or Golang ship with. Though Java 8 is trying to bring out some good stuff, it still is verbose, which means that to perform even small tasks, we end up writing a lot (seriously, a lot) of code. It also ships with `NullPointerException`, which I'm pretty sure scares you and your customers.

Let's dive deeper into the problems in the Java-Android world, which are much worse.

- Android developers are stuck between Java 6 and 7. This means that we don't have features like Lambdas, higher